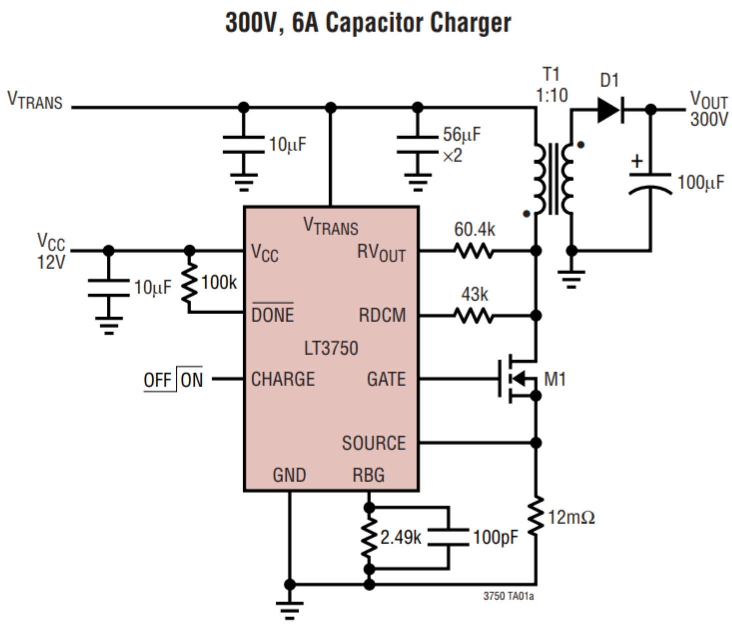
LT3750



Please walk through the datasheet **LT3750** to answer the following questions:

Google: **LT3750 datasheet**

Q1. What is the primary application of LT3750?

The LT3750 is a flyback converter designed to rapidly charge large capacitors to a user-adjustable target voltage. A typical application can charge a 100µF capacitor to 300V in less than 300ms.

Q2. What is the input voltage range of LT3750?

$3\text{ V} < V_{\text{in}} < 24\text{ V}$

Q3. What is the typical switching frequency of LT3750?

$100\text{ kHz} < f_{\text{sw}} < 300\text{ kHz}$

Q4. If we want to implement this circuit, what are some recommended part numbers for the transistor, transformer, and output diode?

Table 1. Recommended Transformers

MANUFACTURER	PART NUMBER	SIZE L × W × H (mm)	MAXIMUM I_{PRI} (A)	L_{PRI} (μH)	TURNS RATIO (PRI:SEC)
TDK (www.tdk.com)	DCT15EFD-U44S003	22.5 × 16.5 × 8.5	5	10	1:10
	DCT20EFD-U32S003	30 × 22 × 12	10	10	1:10
Sumida (www.sumida.com)	C8118 Rev P1	21 × 14 × 8	3	10	1:10
	C8117 Rev P1	23 × 18.6 × 10.8	5	10	1:10
	C8119 Rev P1	32.3 × 27 × 14	10	10	1:10
Midcom (www.midcom.com)	32050	23.1 × 18 × 9.4	3	10	1:10
	32051	28.7 × 22 × 11.4	5	10	1:10
	32052	28.7 × 22 × 11.4	10	10	1:10
Coilcraft (www.coilcraft.com)	DA2032-AL	17.2 × 22 × 8.9	3	10	1:10
	DA2033-AL	17.4 × 24.1 × 10.2	5	10	1:10
	DA2034-AL	20.6 × 30 × 11.3	10	10	1:10

Table 2. Recommended Output Diodes

MANUFACTURER	PART NUMBER	I_{DC} (A)	PEAK REPETITIVE REVERSE VOLTAGE (V)	PACKAGE
Diodes Inc. (www.diodes.com)	MURS140	1	400	SMB
	MURS160	1	600	SMB
	ES2G	2	400	SMB
	US1M	1	1000	SMA
Philips (www.semiconductors.philips.com)	BYD147	1	400	SOD87
	BYD167	1	500	SOD87

Table 4. Recommended NMOS Transisitors

MANUFACTURER	PART NUMBER	I_D (A)	$V_{DS(MAX)}$ (V)	$V_{GS(MAX)}$ (V)	$R_{DS(ON)}$ (mΩ)	PACKAGE
Philips Semiconductor (www.semiconductors.philips.com)	PHM21NQ15T	22.2	150	20	55	HVSON8
	PHK12NQ10T	11.6	100	20	28	SO-8
	PHT6NQ10Y	6.5	100	20	90	SOT223
	PSMN038-100K	6.3	100	20	38	SO-8
International Rectifier (www.irf.com)	IRF7488	6.3	80	20	29	SO-8
	IRF7493	9.3	80	20	15	SO-8
	IRF6644	10.3	100	20	10.7	DirectFET

Q5. Redesign the LT3750 if $V_{\text{out}} = 200V$ is required. (Guide: There are 2 resistors to determine the output voltage, and for the sake of simplicity neglect the diode voltage)

$$V_{out} = 1.24 \times \frac{R_{VOUT}}{R_{BG}} \times N - V_{Diode}$$

$$V_{out} \sim 1.24 \times \frac{R_{VOUT}}{R_{BG}} \times N$$

$$\left.\begin{array}{r} \mathrm{V_{out}} = 200\ \mathrm{V} \\ \mathrm{R_{BG}} = 2.49\ \mathrm{k\Omega} \\ \mathrm{N} = 10 \end{array}\right\} \Rightarrow\ \mathrm{R_{VOUT}} = \frac{\mathrm{R_{BG}} \times \mathrm{V_{out}}}{1.24\mathrm{N}} = 40.1\ \mathrm{k\Omega}$$

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